In [11]:
```
import pandas as pd

funnel = pd.read_excel(
    r"C:\Users\nasee\Downloads\Week 4 Dataset(1).xlsx",
    sheet_name="Funnel Data"
)
```

In [12]:
```
segment_ids = [
    "KC-0007",
    "KC-0017",
    "KC-0033",
    "KC-0047",
    "KC-0061",
    "KC-0071",
    "KC-0075"
]
```

In [13]:
```
segment_funnel = funnel[
    funnel["user_id"].isin(segment_ids)
]

segment_funnel
```

Out[13]:

	user_id	account_created	profile_started	profile_completed	job_viewed	application
6	KC-0007	Yes	Yes	Yes	Yes	
16	KC-0017	Yes	Yes	Yes	Yes	
32	KC-0033	Yes	Yes	Yes	Yes	
46	KC-0047	Yes	Yes	No	No	
60	KC-0061	Yes	Yes	Yes	Yes	
70	KC-0071	Yes	Yes	Yes	Yes	
74	KC-0075	Yes	Yes	Yes	Yes	

In [14]:
```
len(segment_funnel)
```

Out[14]:
```
7
```

In [15]:
```
segment_funnel["user_id"]
```

Out[15]:
```
6     KC-0007
16    KC-0017
32    KC-0033
46    KC-0047
60    KC-0061
70    KC-0071
74    KC-0075
Name: user_id, dtype: str
```

In [16]:
```
stages = [
    "account_created",
    "profile_started",
    "profile_completed",
    "job_viewed",
    "application_submitted",
    "employer_response",
    "offer_received",
    "gig_started"
]
```

In [17]:
```
stage_counts = []

for stage in stages:
    count = (segment_funnel[stage] == "Yes").sum()
    stage_counts.append(count)

stage_counts
```

Out[17]:
```
[np.int64(7),
 np.int64(7),
 np.int64(6),
 np.int64(6),
 np.int64(3),
 np.int64(1),
 np.int64(1),
 np.int64(1)]
```

In [18]:
```
funnel_summary = pd.DataFrame({
    "stage": stages,
    "count": stage_counts
})

funnel_summary
```

Out[18]:

	stage	count
0	account_created	7
1	profile_started	7
2	profile_completed	6
3	job_viewed	6
4	application_submitted	3
5	employer_response	1
6	offer_received	1
7	gig_started	1

In [19]:

```
funnel_summary["step_conversion"] = (
    funnel_summary["count"] /
    funnel_summary["count"].shift(1)
)

funnel_summary["step_conversion"] = (
    funnel_summary["step_conversion"] * 100
).round(1)

funnel_summary
```

Out[19]:

	stage	count	step_conversion
0	account_created	7	NaN
1	profile_started	7	100.0
2	profile_completed	6	85.7
3	job_viewed	6	100.0
4	application_submitted	3	50.0
5	employer_response	1	33.3
6	offer_received	1	100.0
7	gig_started	1	100.0

In [20]:

```
funnel_summary["absolute_loss"] = (
    funnel_summary["count"].shift(1)
    - funnel_summary["count"]
)

funnel_summary
```

Out[20]:

	stage	count	step_conversion	absolute_loss
0	account_created	7	NaN	NaN
1	profile_started	7	100.0	0.0
2	profile_completed	6	85.7	1.0
3	job_viewed	6	100.0	0.0
4	application_submitted	3	50.0	3.0
5	employer_response	1	33.3	2.0
6	offer_received	1	100.0	0.0
7	gig_started	1	100.0	0.0

In [21]:

```
funnel_summary["relative_loss"] = (
    funnel_summary["absolute_loss"]
    / funnel_summary["count"].shift(1)
)

funnel_summary["relative_loss"] = (
    funnel_summary["relative_loss"] * 100
).round(1)

funnel_summary
```

Out[21]:

	stage	count	step_conversion	absolute_loss	relative_loss
0	account_created	7	NaN	NaN	NaN
1	profile_started	7	100.0	0.0	0.0
2	profile_completed	6	85.7	1.0	14.3
3	job_viewed	6	100.0	0.0	0.0
4	application_submitted	3	50.0	3.0	50.0
5	employer_response	1	33.3	2.0	66.7
6	offer_received	1	100.0	0.0	0.0
7	gig_started	1	100.0	0.0	0.0

In [22]:

```
funnel_summary.sort_values(
    "absolute_loss",
    ascending=False
)
```

Out[22]:

	stage	count	step_conversion	absolute_loss	relative_loss
4	application_submitted	3	50.0	3.0	50.0
5	employer_response	1	33.3	2.0	66.7
2	profile_completed	6	85.7	1.0	14.3
3	job_viewed	6	100.0	0.0	0.0
1	profile_started	7	100.0	0.0	0.0
6	offer_received	1	100.0	0.0	0.0
7	gig_started	1	100.0	0.0	0.0
0	account_created	7	NaN	NaN	NaN

In [23]:
```
funnel_summary.sort_values(
    "relative_loss",
    ascending=False
)
```

Out[23]:

	stage	count	step_conversion	absolute_loss	relative_loss
5	employer_response	1	33.3	2.0	66.7
4	application_submitted	3	50.0	3.0	50.0
2	profile_completed	6	85.7	1.0	14.3
3	job_viewed	6	100.0	0.0	0.0
1	profile_started	7	100.0	0.0	0.0
6	offer_received	1	100.0	0.0	0.0
7	gig_started	1	100.0	0.0	0.0
0	account_created	7	NaN	NaN	NaN

RETENTION

In [24]:
```
weekly = pd.read_excel(
    r"C:\Users\nasee\Downloads\Week 4 Dataset(1).xlsx",
    sheet_name="Weekly Activity"
)
```

In [25]:
```
segment_weekly = weekly[
    weekly["user_id"].isin(segment_ids)
]

segment_weekly.head()
```

Out[25]:

	user_id	week_number	active_flag	jobs_viewed	applications	employer_responses
24	KC-0007	1	Yes	6	1	0
25	KC-0007	2	No	0	0	0
26	KC-0007	3	Yes	3	2	2
27	KC-0007	4	Yes	4	3	1
64	KC-0017	1	No	0	0	0

In [26]:
```
week1_cohort = segment_weekly[
    (segment_weekly["week_number"] == 1)
    &
    (segment_weekly["active_flag"] == "Yes")
]

week1_cohort
```

Out[26]:

	user_id	week_number	active_flag	jobs_viewed	applications	employer_responses
24	KC-0007	1	Yes	6	1	0
128	KC-0033	1	Yes	1	0	0
240	KC-0061	1	Yes	2	0	0
296	KC-0075	1	Yes	7	1	0

In [27]:
```
week1_ids = week1_cohort["user_id"].unique()

print(week1_ids)
```

```
<ArrowStringArray>
['KC-0007', 'KC-0033', 'KC-0061', 'KC-0075']
Length: 4, dtype: str
```

In [28]:
```
len(week1_ids)
```

Out[28]: 4

In [29]:
```
cohort_size = len(week1_ids)

for week in [1, 2, 3, 4]:

    active_users = segment_weekly[
```

```
        (segment_weekly["week_number"] == week)
        &
        (segment_weekly["active_flag"] == "Yes")
        &
        (segment_weekly["user_id"].isin(week1_ids))
    ]["user_id"].nunique()

    retention = active_users / cohort_size * 100

    print(
        f"Week {week}: "
        f"{active_users} users, "
        f"{retention:.1f}% retention"
    )
```

```
Week 1: 4 users, 100.0% retention
Week 2: 1 users, 25.0% retention
Week 3: 3 users, 75.0% retention
Week 4: 2 users, 50.0% retention
```

In [30]:

```
retention_data = []

for week in [1, 2, 3, 4]:

    active_users = segment_weekly[
        (segment_weekly["week_number"] == week)
        &
        (segment_weekly["active_flag"] == "Yes")
        &
        (segment_weekly["user_id"].isin(week1_ids))
    ]["user_id"].nunique()

    retention = round(
        active_users / cohort_size * 100,
        1
    )

    retention_data.append(
        [week, active_users, retention]
    )

retention_table = pd.DataFrame(
    retention_data,
    columns=[
        "Week",
        "Active Users",
        "Retention %"
    ]
)

retention_table
```

Out[30]:

	Week	Active Users	Retention %
0	1	4	100.0
1	2	1	25.0
2	3	3	75.0
3	4	2	50.0

In [31]:
```
all_week1_ids = weekly[
    (weekly["week_number"] == 1)
    &
    (weekly["active_flag"] == "Yes")
]["user_id"].unique()

all_cohort_size = len(all_week1_ids)
```

In [32]:
```
all_retention = []

for week in [1,2,3,4]:

    active_users = weekly[
        (weekly["week_number"] == week)
        &
        (weekly["active_flag"] == "Yes")
        &
        (weekly["user_id"].isin(all_week1_ids))
    ]["user_id"].nunique()

    retention = round(
        active_users /
        all_cohort_size * 100,
        1
    )

    all_retention.append(
        [week, active_users, retention]
    )

all_retention_table = pd.DataFrame(
    all_retention,
    columns=[
        "Week",
        "All Users Active",
        "All User Retention %"
    ]
)

all_retention_table
```

Out[32]:

	Week	All Users Active	All User Retention %
0	1	176	100.0
1	2	100	56.8
2	3	93	52.8
3	4	77	43.8

In [33]:

```
comparison = retention_table.merge(
    all_retention_table,
    on="Week"
)

comparison
```

Out[33]:

	Week	Active Users	Retention %	All Users Active	All User Retention %
0	1	4	100.0	176	100.0
1	2	1	25.0	100	56.8
2	3	3	75.0	93	52.8
3	4	2	50.0	77	43.8